

NYCU-EE IC LAB – 2024 Fall

Lab12 Exercise

Design: Train Tour APRII

Data Preparation

1. Extract test data from TA's directory:
`% tar xvf ~iclabTA01/Lab12.tar`
2. The extracted LAB directory contains:
 - a. 00_TESTBED
 - b. 05_APR

- c. 06_POST
- d. 09_SUBMIT

Description

In this lab you will finish the **backend flow (APR)** for Lab03(TA provided netlist files) and use the **IR drop and Power Analysis** tool for your layout.

Inputs and Outputs

Input signal	Bit Width	Description	
clk	1	Global clock signal. All signals are sampled on the Positive edge of the global clock.	
rst_n	1	Global reset signal. This signal is active LOW.	
in_valid	1	High when the input is valid.	
tetrominoes	3	Denote the type of the tetromino. Please refer to the Introduction part to check the index of different type of tetromino.	
position	3	Mark the location of the leftmost block of the tetromino.	
		tetrominoes	position
		0, 3, 5, 6	3'd0~3'd4
		1	3'd0~3'd5
		2	3'd0~3'd2
		4, 7	3'd0~3'd3

Output Signal	Bit Width	Description
score_valid	1	HIGH when the score is valid. The score must be valid for every (tetrominoes, position) set in 1 game round.

score	4	The score the player earns in each round of game.
tetris_valid	1	HIGH when the tetris is valid. Must be HIGH at the end of the current round, which is exactly the same cycle of the last score_valid of the current game round.
tetris	12 x 6	A 12x6 unpacked array. Record the tetris map of the current round.
fail	1	A flag that indicates the current round has terminated early. 1: the player loses and the game terminates early. 0: the player wins.

Layout Specifications

1. CHIP.sdc period is **fixed to 3.5ns** input/output delay is **fixed to 1.75ns**
2. Core power pad and io power pad
 - a. At least one pair at each side.
3. Floorplanning
 - a. Core size:
 - ✓ Defined by you
 - b. Core to IO boundary:
 - ✓ Each side must be larger than **100**
4. Power Planning
 - a. Core Ring
 - ✓ Top & Bottom: metal layer must be odd and width is **fixed to 9**.
 - ✓ Left & Right: metal layer must be even and width is **fixed to 9**.
 - ✓ Each side must be **wire group, interleaving**, and **at least 4** pairs.
 - b. Stripes
 - ✓ Vertical: metal layer must be even and width is defined by you.
 - ✓ Horizontal: metal layer must be odd and width is defined by you.
 - ✓ Number of pairs is defined by you
 - c. Timing Analysis
 - ✓ Timing Slack:
 - NO negative slacks** after setup/hold time analysis (Post-Route stage)
 - ✓ Design Rule Violation (DRV)
 - The DRV of (fanout, cap, tran) should be all **0** after setup/hold time analysis
 - d. Design Verification Result
 - ✓ LVS : **No LVS violations** after “verify connectivity”
 - ✓ DRC : **No DRC violations** after “verify DRC”
 - ✓ **Note: Remember to check DRC / LVS again after placing the fillers.**
Although in normal cases, if DRC and LVS are verified after nanoRoute, no more DRC / LVS will be produced during postRoute optimization and adding core filler. However, some special cases produce the further DRC and LVS.
For example: inserting a filler that is isolated from all other cells and power lines, then open LVS occurs. Thus, be sure to verify all the specs above after performing all APR steps.
 - e. Rail Analysis:
 - ✓ VCC Threshold set to **1.7**
 - ✓ GND Threshold set to **0.1**
 - ✓ No IR drop is allowed larger than **2mV**

Note

1. Complete **CHIP.io** and **CHIP_SHELL.v**
2. Do all the flow as in APRI (Lab11) with Layout Specification above
3. Run Power Analysis (Setup & Run)
4. Run Rail Analysis
 - Set PG Library Mode
 - Generate PG Library
 - Setup & Run Rail Analysis
5. Observe the IR Drop to analyze whether IR drop is within **2mV**.
6. Please submit your files under 09_SUBMIT and **a report to the E3 system** before 12:00 at noon on December. 09:
 - **The report should include screenshots and brief explanations. All necessary information is included in “Lab12_report_example.docx”.**
Your report need to be named as “Lab12_report_iclabXXX.pdf”.
 - If uploaded files **violate the naming rule**, you will get **5 deduct point**.

```
[Exercise/09_SUBMIT]% ./01_submit
```

After that, you should check the following files under 09_SUBMIT/Lab12_iclabXXX/

- i. **CHIP_iclabXXX.inn**
- ii. **CHIP_iclabXXX.inn.dat** (this one is a directory)
- iii. **CHIP_SHELL_iclabXXX.v**
- iv. **CHIP_iclabXXX.io**

If you miss any files on the list, you will fail this lab.

Then use the command like the figure below to check the files are uploaded or not

```
[Exercise/09_SUBMIT]% ./02_check 1st_demo
```

7. **Self-Verify** APR Result!!! Remember to **restore** and check the **checklist**.

Grading Policy

- **You should meet all the Layout Specification and File Specification above, pass post simulation without any timing violation, and within tolerable IR drop to pass the demo.**
- **Report: 100 points**
- **2de pass grade: 70 points**

